

# HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

Anonymous Authors

Affiliation(s) omitted for blind review

## ABSTRACT

Tiny-object detection in high-resolution aerial and maritime imagery must contend with vast, redundant backgrounds and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a dual-evidence framework featuring semantic-spectral routing, coverage-constrained spatial optimization, and lightweight downstream detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. To promote recall-safe routing, an object-level soft-coverage loss directly optimizes multi-cell joint instance coverage across ground-truth objects. To recover downstream efficiency, an inverted residual decoupled head (ISPP) is integrated to compress parameters by  $> 27\%$  with only fractional FLOPs overhead. Experiments show that HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, at  $> 27\%$  fewer parameters and real-time speed, making it well suited to resource-constrained deployment.

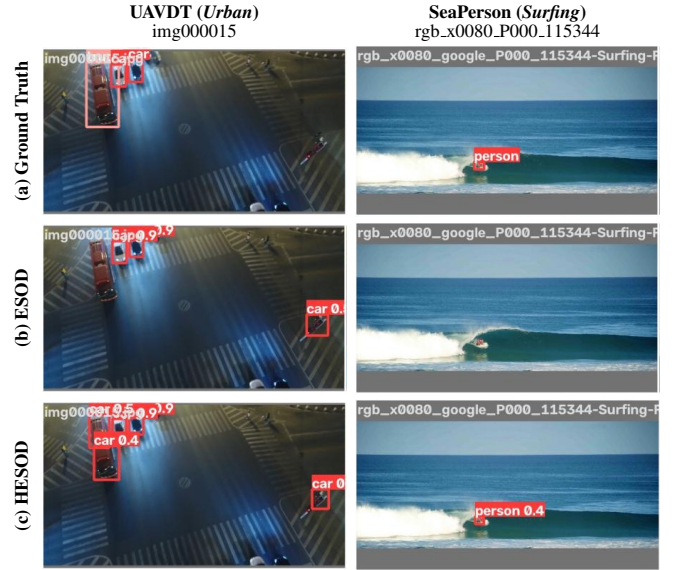
**Index Terms**— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

## 1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ( $< 16 \times 16$  px) amid massive uniform backgrounds ( $> 70\%$ – $80\%$ ) [4, 5]. Evaluating standard dense deep detectors across multi-megapixel grids ( $1280 \times 1280$  to  $2048 \times 2048$ ) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* dynamically routes feature processing strictly to prospective foreground regions [4, 6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap directly after a shallow stem, slicing features into candidate bounding patches (AdaSlicer) and bypassing background across downstream stages.

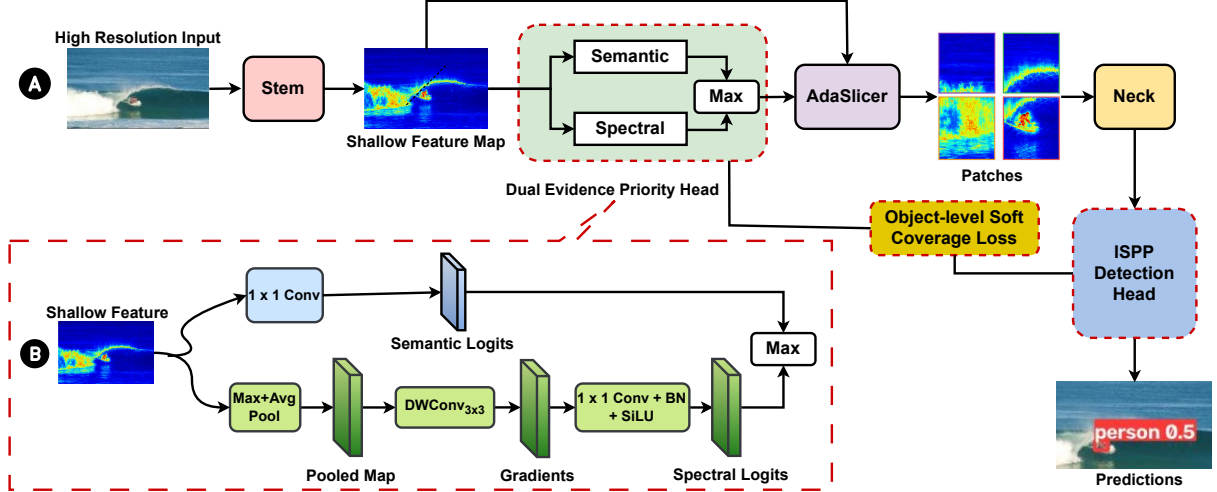
Crucially, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative* across downstream stages [4]. Existing detectors rely exclusively on single semantic objectness with independent per-cell BCE. Under extreme downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we



**Fig. 1.** Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

observe that tiny targets retain prominent high-frequency spatial gradients and spectral saliency even when deep semantics fade [8–10]. We reveal that naive selective inference suffers from acute semantic collapse under asymmetric pruning risks, and propose **HESOD** (High-Resolution Efficient Small Object Detection) to establish recall-safe, evidence-preserving sparse routing. Our key contributions are:

1. **Dual-Evidence Spatial Routing:** We uncover single-semantic routers’ vulnerability to feature collapse, coupling semantic objectness with constant-width gradient filtering for scale-robust structural priors independent of backbone capacity.
2. **Evidence Preservation and Soft Coverage:** We formulate a monotonic max-fusion rule preventing multi-cue evidence dilution, paired with an object-level noisy-OR soft-coverage loss directly optimizing joint instance recall under irreversible pruning.
3. **Architectural Compute Compensation:** We co-design downstream detection with an inverted residual decoupled head [11] as a dedicated compute compensator, amortizing candidate evaluation overhead while slashing detector parameters by  $> 27\%$ .



**Fig. 2.** Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Dashed boxes highlight our proposed contributions over ESOD.

## 2. RELATED WORK

### 2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ( $> 70\%$ – $80\%$ ) [1–3, 12]. Classical approaches utilize density-guided cropping (ClusDet [13], DMNet [14]) or scale-adaptive label assignment (NWD [15], RFLA [16]). Recently, ResBiDet [17] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

### 2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7], and sparse location activation [18]). Recently, coarse-to-fine detectors such as ESOD [4] and OSDT [19] predict early spatial priority heatmaps to extract candidate patches. However, existing selectors rely exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

### 2.3. Spectral Saliency and Frequency Methods

Frequency-domain representations offer rich structural priors for micro targets, which retain sharp high-frequency edge gradients even when photometric semantics fade [8–10]. While recent methods explore dense frequency transforms [8] or detail compensation across multi-scale feature pyramids [20], they incur substantial downstream latency. In sharp contrast, HESOD repurposes high-frequency gradient saliency upstream as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional

Laplacian and Sobel gradient filtering at constant width, independent of backbone channel dimension  $C$ .

## 3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2(a). From input  $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ , a shallow stem extracts dense features  $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$ . The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map  $\mathbf{S} \in [0, 1]^{H_s \times W_s}$ ; and (ii) an AdaSlicer module dynamically clustering candidate foreground cells into bounding patches  $\mathcal{P}$  from  $\mathbf{F}$ . Downstream backbone stages and lightweight decoupled heads evaluate only retained patches  $\mathcal{P}$  under soft-coverage supervision (§3.2, §3.3).

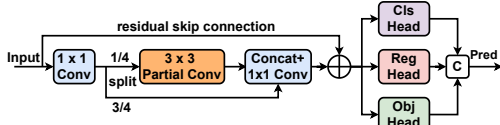
### 3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let  $\mathbf{f}_i \in \mathbb{R}^C$  denote the feature vector at spatial location  $i$  in  $\mathbf{F}$ . The semantic branch predicts class-specific logits via a  $1 \times 1$  convolution:  $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1 \times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$ . In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[ \max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (1)$$

where  $[\cdot \parallel \cdot]$  denotes channel concatenation. Compressing  $\mathbf{F}$  to 2 channels retains peak edge activation and background energy, after which a learnable depthwise  $3 \times 3$  convolution bank—initialized with discrete Laplacian and directional Sobel operators—filters  $\mathbf{F}_{\text{pooled}}$  into 6 structural channels at  $\mathcal{O}(H_s W_s)$  cost independent of width  $C$ . In the 2D spatial frequency domain, the discrete Laplacian is approximately isotropic in the low-frequency limit, with  $|H(\omega)| \propto \|\omega\|^2$ , naturally amplifying edge transitions. Because micro targets ( $< 16^2$  px) manifest as localized high-frequency spatial impulses whose gradient energy resists downsampling [10], this branch supplies a constant-bandwidth prior independent of deep semantic capacity. A  $1 \times 1$  projection maps features to  $N_c$ , yielding spectral logits  $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$ .



**Fig. 3.** Inverted Residual Decoupled Head (ISPP) with Partial Convolution (© denotes channel concatenation).

To prevent confidence dilution on faint targets, evidence fusion enforces *evidence preservation*. Linear combinations risk destructive interference: an unconfident or negative activation in one branch can drag down a confident signal in the other below the routing threshold. Instead, we formulate a parameter-free elementwise maximum:

$$z_{i,c} = \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}}), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

This operation functions as an evidence-preserving OR gate: since  $z_{i,c} \geq z_{i,c}^{\text{sem}}$  and  $z_{i,c} \geq z_{i,c}^{\text{spec}}$ , neither branch can ever suppress genuine candidate cues. Cells with priority  $s_i > \tau$  are clustered into bounding patches  $\mathcal{P}$  for downstream detection.

### 3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let  $\mathcal{N}(j)$  denote the set of selector cells overlapping ground-truth box  $j \in \{1, \dots, N_{\text{gt}}\}$ . We formulate a noisy-OR differentiable coverage surrogate and its weighted instance loss:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i), \quad \ell_j = -w_j \log(p_j^{\text{cover}} + \epsilon), \quad (3)$$

where  $w_j = \text{clip}(4/a_j, 1, 5)$  is a scale-adaptive weight based on bounding box area  $a_j$  in cells, penalizing misses on micro targets ( $a_j < 4$ ). For any cell  $i \in \mathcal{N}(j)$ , the gradient of  $\ell_j$  with respect to cell score  $s_i$  is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \quad (4)$$

When neighboring candidate cells have low scores ( $s_k \rightarrow 0$ ), the product term approaches 1, generating a strong gradient to boost cell  $i$  and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances:  $\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$ . The multi-task training objective is  $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}})$ , where  $\mathcal{L}_{\text{BCE}}$  is the per-cell selector BCE over scores  $s_i$ , with  $\lambda_{\text{sel}} = 0.2$  and  $\lambda_{\text{cov}} = 0.5$ .

### 3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an inverted residual partial-convolution (ISPP) decoupled head [11] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels of width  $C_1$  to  $2C_1$ , partitioning them into an active branch (0.25 ratio) processed by a  $3 \times 3$  Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled  $1 \times 1$  heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

**Table 1.** Benchmark results on UAVDT and SeaPerson.

| Method                                   | mAP@.5         | Params (M)    | GFLOPs         | Total Recall    | BPR            | FPS          |
|------------------------------------------|----------------|---------------|----------------|-----------------|----------------|--------------|
| <b>UAVDT Benchmark</b> (1280 × 1280)     |                |               |                |                 |                |              |
| Faster R-CNN [21]                        | 0.336          | 43.3          | 772.5          | 79.68%          | –              | 18.8         |
| RetinaNet [22]                           | 0.296          | 36.4          | 360.9          | 83.44%          | –              | 19.8         |
| QueryDet <sup>†</sup> [6]                | 0.257          | 39.3          | – <sup>†</sup> | 41.89%          | –              | 21.1         |
| ESOD [4]                                 | <u>0.385</u>   | <u>35.9</u>   | <b>68.2</b>    | 85.17%          | <u>0.884</u>   | <b>117.8</b> |
| <b>HESOD (Ours)</b>                      | <b>0.394</b>   | <b>26.0</b>   | 74.9           | <b>88.83%</b>   | <b>0.940</b>   | <u>106.1</u> |
| Δ vs. ESOD                               | <b>+0.9 pp</b> | <b>-27.5%</b> | +9.8%          | <b>+3.66 pp</b> | <b>+5.6 pp</b> | -9.9%        |
| <b>SeaPerson Benchmark</b> (2048 × 2048) |                |               |                |                 |                |              |
| Faster R-CNN [21]                        | 0.551          | 43.3          | 1546.8         | 67.27%          | –              | 23.1         |
| RetinaNet [22]                           | 0.473          | 36.4          | 942.7          | 65.98%          | –              | 28.0         |
| QueryDet <sup>†</sup> [6]                | 0.627          | 39.3          | – <sup>†</sup> | 61.56%          | –              | 14.0         |
| ESOD [4]                                 | <u>0.750</u>   | <u>35.8</u>   | <b>202.4</b>   | 84.42%          | <u>0.947</u>   | <b>85.7</b>  |
| <b>HESOD (Ours)</b>                      | <b>0.773</b>   | <b>25.9</b>   | <u>208.5</u>   | <b>87.75%</b>   | <b>0.991</b>   | <u>80.6</u>  |
| Δ vs. ESOD                               | <b>+2.3 pp</b> | <b>-27.6%</b> | +3.0%          | <b>+3.33 pp</b> | <b>+4.4 pp</b> | -6.0%        |

BPR: Bounding Patch Recall. **Bold**: best; underline: second-best (selective models).

<sup>†</sup>QueryDet uses CSQ (dynamic sparse FLOPs unamenable to static profiling; dense query counterparts: 973.5 / 2539.0 GFLOPs at 12.5 / 10.6 FPS on UAVDT / SeaPerson).

**Two-Stage Training Protocol.** To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1 (Selector Convergence)*: Train the Dual-Max selector with a coupled head under  $\mathcal{L}_{\text{cover}}$ , establishing a well-calibrated priority heatmap; and (ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak coverage (0.940 BPR), maintaining competitive real-time FPS with only fractional overhead vs. ESOD.

## 4. EXPERIMENTS

### 4.1. Experimental Setup

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280 × 1280 px, urban vehicle surveillance across 3 classes, 19.9% extremely tiny targets < 16<sup>2</sup> px) and **SeaPerson** [3, 12] (2048 × 2048 px, maritime search-and-rescue with > 85% targets < 32 px). Both benchmarks use full-resolution images without artificial cropping.

Selective detectors (ESOD family and our ablations) employ YOLOv5m backbones in PyTorch 2.8 trained for 50 epochs under selective protocols [4]. Dense and query baselines (Faster R-CNN, RetinaNet, QueryDet) retain canonical ResNet-50-FPN backbones, evaluated on identical test splits and resolutions. All models are benchmarked on an NVIDIA RTX 5090 GPU. Metrics include COCO mAP@.5 and mAP@.5 : .95, scale-stratified recall across four bins (Very Tiny < 16<sup>2</sup>, Tiny 16<sup>2</sup>–32<sup>2</sup>, Small 32<sup>2</sup>–96<sup>2</sup>, Med/Large > 96<sup>2</sup> px), and Bounding Patch Recall (BPR), measuring the fraction of ground-truth targets covered (> 50% overlap) by selected patches.

### 4.2. Main Benchmark Comparisons

As shown in Table 1, traditional dense detectors suffer from severe efficiency bottlenecks on high-resolution images: Faster R-CNN only achieves 18.8/23.1 FPS on UAVDT/SeaPerson with GFLOPs as high as 772.5/1546.8; QueryDet [6] is limited by its dynamic query mechanism, with total recall dropping to only 41.89% on UAVDT. HESOD outperforms all baselines: on UAVDT, mAP@.5 is 5.8, 9.8, and 13.7 pp higher than Faster R-CNN, RetinaNet, and

**Table 2.** Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

| Configuration                                   | Detection Accuracy and Computational Efficiency |                |              |               |              |              | Physical-Scale Recall Breakdown |                         |                          |                         |                 |
|-------------------------------------------------|-------------------------------------------------|----------------|--------------|---------------|--------------|--------------|---------------------------------|-------------------------|--------------------------|-------------------------|-----------------|
|                                                 | mAP<br>@.5                                      | mAP<br>@.5:.95 | BPR          | Params<br>(M) | GFLOPs       | FPS          | Very Tiny<br>( $< 16^2$ )       | Tiny<br>( $16^2-32^2$ ) | Small<br>( $32^2-96^2$ ) | Med/Lrg<br>( $> 96^2$ ) | Total<br>Recall |
| <b>UAVDT Benchmark</b> (1280 $\times$ 1280)     |                                                 |                |              |               |              |              | (74,375)                        | (206,015)               | (86,282)                 | (7,325)                 | (373,997)       |
| (1) ESOD Baseline*                              | 0.385                                           | 0.214          | 0.884        | 35.9          | <b>68.2</b>  | <b>117.8</b> | 79.43%                          | 84.21%                  | 94.54%                   | 59.78%                  | 85.17%          |
| (2) Semantic-only                               | 0.384                                           | <u>0.217</u>   | 0.906        | 35.9          | 75.0         | <u>113.8</u> | 79.13%                          | 83.91%                  | 94.00%                   | 59.95%                  | 84.82%          |
| (3) Spectral-only                               | 0.387                                           | 0.209          | <u>0.942</u> | 36.0          | 93.7         | 102.3        | 82.43%                          | 88.72%                  | 96.66%                   | <u>66.80%</u>           | 88.87%          |
| (4) (3) + Channel-Pooled                        | <u>0.394</u>                                    | 0.209          | <b>0.963</b> | 35.9          | 99.3         | 97.6         | <b>86.88%</b>                   | <b>92.23%</b>           | <b>97.74%</b>            | 66.43%                  | <b>91.93%</b>   |
| (5) Dual-Concat                                 | 0.371                                           | 0.205          | 0.919        | 35.9          | 83.9         | 107.2        | 79.09%                          | 85.85%                  | 95.85%                   | 62.66%                  | 86.35%          |
| (6) Dual-Max                                    | <b>0.395</b>                                    | <b>0.218</b>   | 0.940        | 35.9          | 90.1         | 102.3        | <u>85.69%</u>                   | <u>89.97%</u>           | <u>97.37%</u>            | 65.99%                  | <u>90.36%</u>   |
| (7) <b>HESOD (Ours)</b>                         | <u>0.394</u>                                    | 0.215          | 0.940        | <b>26.0</b>   | <u>74.9</u>  | 106.1        | 80.76%                          | 88.98%                  | 97.11%                   | <b>69.16%</b>           | 88.83%          |
| <b>SeaPerson Benchmark</b> (2048 $\times$ 2048) |                                                 |                |              |               |              |              | (82,417)                        | (174,816)               | (42,987)                 | (155)                   | (300,375)       |
| (1) ESOD Baseline*                              | 0.750                                           | 0.320          | 0.947        | 35.8          | <b>202.4</b> | <b>85.7</b>  | 74.03%                          | 86.78%                  | 94.74%                   | 79.35%                  | 84.42%          |
| (2) Semantic-only                               | 0.769                                           | 0.325          | 0.991        | 35.8          | 266.8        | 73.5         | 75.49%                          | 91.57%                  | 95.71%                   | 81.94%                  | 87.75%          |
| (3) Spectral-only                               | 0.770                                           | <u>0.327</u>   | <b>0.992</b> | 35.8          | 267.4        | 71.7         | 75.23%                          | <u>91.69%</u>           | <b>95.89%</b>            | <u>86.45%</u>           | 87.77%          |
| (4) (3) + Channel-Pooled                        | 0.767                                           | 0.324          | 0.988        | 35.8          | 263.4        | 72.7         | <b>76.13%</b>                   | 91.36%                  | 95.50%                   | <b>87.10%</b>           | 87.77%          |
| (5) Dual-Concat                                 | 0.772                                           | 0.326          | 0.986        | 35.8          | 281.2        | 69.8         | <u>76.00%</u>                   | 91.50%                  | 95.68%                   | 80.00%                  | <u>87.84%</u>   |
| (6) Dual-Max                                    | <b>0.778</b>                                    | <b>0.330</b>   | <u>0.991</u> | 35.8          | 255.1        | 73.8         | 75.65%                          | <b>92.06%</b>           | <u>95.85%</u>            | 81.94%                  | <b>88.10%</b>   |
| (7) <b>HESOD (Ours)</b>                         | <u>0.773</u>                                    | <u>0.327</u>   | <u>0.991</u> | <b>25.9</b>   | <u>208.5</u> | 80.6         | 75.71%                          | 91.47%                  | 95.72%                   | 80.65%                  | 87.75%          |

\*Baseline uses BCE only; others add  $\mathcal{L}_{\text{cover}}$  on top. Parenthetical values denote ground-truth counts per bin. One-to-one matching at conf  $\geq 0.001$ , IoU  $\geq 0.5$ . **Bold**: best; underline: second-best.

QueryDet respectively at  $5.0\times-5.6\times$  higher FPS (106.1 vs. 18.8–21.1); on SeaPerson, mAP@.5 exceeds all baselines by 14.6~30.0 pp at  $2.9\times-5.8\times$  higher FPS, with Very Tiny recall reaching 75.71% (+27.0 pp over QueryDet).

Against the same-architecture selective baseline ESOD [4]: on UAVDT, mAP@.5 increases from 0.385 to 0.394 (+0.9 pp), BPR from 0.884 to 0.940 (+5.6 pp), total recall by +3.66 pp, with bus class recall jumping from 67.7% to 81.4% (+13.7 pp); parameters are reduced from 35.9M to 26.0M (−27.5%) at 106.1 FPS. On SeaPerson, mAP@.5 increases from 0.750 to 0.773 (+2.3 pp), BPR reaches 0.991 (+4.4 pp), total recall increases by +3.33 pp, successfully recalling more than 10,000 micro maritime targets missed by ESOD; GFLOPs only increase by 3.0%, parameters are compressed by 27.6% at 80.6 FPS. The modest GFLOPs overhead stems from the selector routing more genuine target regions downstream rather than computational redundancy; HESOD controls this within 10% via the lightweight head design while ensuring >99% target coverage.

### 4.3. Ablation Studies and Scale Analysis

We conduct controlled ablation experiments on the ESOD baseline by adding components incrementally, with results in Table 2.

**Dual-Evidence Fusion.** Using only the semantic branch yields 0.906 BPR and 84.82% total recall on UAVDT, showing limited improvement due to semantic feature collapse during downsampling; the spectral-only branch achieves stronger BPR of 0.942 and 88.87% total recall, as gradient features remain robust on micro targets. Comparing fusion strategies, linear concatenation (Arm 5) achieves only 0.371 mAP@.5—lower than either single branch—because weak semantic responses introduce destructive interference that suppresses confident spectral activations. In contrast, element-wise maximum (Dual-Max, Arm 6) implements an evidence-preserving OR gate, achieving 0.395/0.778 mAP@.5 and 0.940/0.991 BPR on UAVDT/SeaPerson, the best among all single-stage selector configurations.

**Soft-Coverage Loss.** The ESOD baseline trained with per-pixel BCE achieves only 0.884 BPR and 79.43% Very Tiny recall on UAVDT. With the proposed  $\mathcal{L}_{\text{cover}}$ , the semantic branch’s Very Tiny recall is flat at 79.13%, while the spectral branch’s rises to 82.43%; under Dual-Max fusion it reaches 85.69% (+6.26 pp). The noisy-OR formulation ensures that when all cells covering a target

have low scores, the coverage gradient boosts at least one cell above the routing threshold, fundamentally reducing unrecoverable missed detections.

**ISPP Head Efficiency.** Without the ISPP head, Dual-Max incurs 90.1 GFLOPs and 102.3 FPS on UAVDT. Adding ISPP reduces GFLOPs by 16.9% to 74.9, boosts FPS to 106.1, and compresses parameters by 27.5%, with mAP@.5 decreasing by only 0.001 and BPR unchanged at 0.940. On SeaPerson, ISPP cuts GFLOPs by 18.2% and parameters by 27.6%, lifting FPS from 73.8 to 80.6 with negligible accuracy loss (0.778  $\rightarrow$  0.773 mAP).

**Channel Pooling.** Without channel pooling, full-width spectral filtering limits batch size to 2 on SeaPerson due to memory constraints, yielding only 0.763 mAP@.5. Compressing features to 2 channels enables batch-8 training, boosting mAP@.5 by +1.5 pp to 0.778, reducing GFLOPs by 8.2%, and further lifting Very Tiny recall by +1.2 pp.

**Error Bottleneck Diagnosis.** Coverage-loss supervision alone reduces selector false negatives from 25.6% to 19.2% on SeaPerson’s semantic branch (BPR 0.947 $\rightarrow$ 0.991), while dual-evidence routing rescues more than 13,600 dropped instances on UAVDT (BPR 0.884  $\rightarrow$  0.940), shifting the remaining bottleneck to downstream regression refinement.

**Latency Profiling.** On an RTX 5090 GPU, stem extraction takes 1.2 ms, selector 1.8 ms, AdaSlicer 0.9 ms, and ISPP 5.5 ms, achieving 106.1/80.6 FPS on UAVDT/SeaPerson respectively, which meets the real-time inference requirements of edge drone platforms.

## 5. CONCLUSION

We presented HESOD, a dual-evidence framework uniting semantic objectness with constant-width spectral saliency via monotonic max-fusion, addressing early semantic collapse on micro targets under selective high-resolution detection. An object-level soft-coverage loss ( $\mathcal{L}_{\text{cover}}$ ) optimizes instance coverage under asymmetric pruning risks, while an inverted residual decoupled head (ISPP) amortizes downstream evaluation overhead. On UAVDT and SeaPerson, HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (total recall +3.66 pp and +3.33 pp), cuts parameters by > 27%, and achieves 106.1/80.6 FPS on an RTX 5090. Dual-evidence routing reaches 0.940/0.991 BPR, shifting the remaining bottleneck to downstream regression refinement and enabling accuracy-efficiency co-design for edge deployment.

## 6. REFERENCES

- [1] Pengfei Zhu, Longyin Wen, Dawei Du, Xiao Bian, Heng Fan, Qinghua Hu, and Haibin Ling, "Detection and tracking meets drones: A benchmark," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7388–7404, 2021.
- [2] Dawei Du, Yuankai Qi, Hongyang Yu, Yifan Yang, Kaiwen Duan, Guorong Li, Weigang Zhang, Qingming Huang, and Ling Shao, "The unmanned aerial vehicle benchmark: Object detection and tracking," in *European Conference on Computer Vision*, 2018, pp. 370–386.
- [3] Xue-Yang Zhang, Guanying Xu, Jin-Yu Fan, and Haibin Ling, "Scale match for tiny person detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2020, pp. 1256–1265.
- [4] Kai Liu, Zhunga Fu, Shan Jin, Zhipeng Chen, Feng Zhou, Rong Jiang, Yonghong Chen, and Jinwei Ye, "ESOD: Efficient small object detection on high-resolution images," *IEEE Transactions on Image Processing*, vol. 34, pp. 183–195, 2025.
- [5] Mahyar Najibi, Bharat Singh, and Larry S. Davis, "Auto-Focus: Efficient multi-scale inference," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 9745–9755.
- [6] Chenhongyi Yang, Zehao Huang, and Naiyan Wang, "Query-Det: Cascaded sparse query for accelerating high-resolution small object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16668–16677.
- [7] Bowei Du, Yanan Huang, Jiahui Chen, and Dong Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13435–13444.
- [8] Hao Sun, Rui Wang, Yufeng Li, Chenhongyi Yang, Zehao Huang, Naiyan Wang, and Lin Zhou, "SET: Spectral enhancement for tiny object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 1–10.
- [9] Fangrui Guo, Junwei Wu, and Quan Zhang, "Dual-domain feature-guided task alignment for enhanced small object detection," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [10] M. Zhao et al., "Frequency-gradient collaborative network for infrared small target detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–14, 2025.
- [11] Xiaocong Xu, Qi Li, Jeng-Shyang Pan, Linfeng Wei, and Xiao-Zhi Gao, "ESOD-YOLO: An enhanced efficient small object detection framework for aerial images," *Computing*, vol. 107, pp. 54, 2025.
- [12] Leon A. Varga, Benjamin Kiefer, Martin Messmer, and Andreas Zell, "SeaDronesSee: A maritime benchmark for detecting humans in open water," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2260–2270.
- [13] Fanyi Yang, Heng Fan, Peng Chu, Erik Blasch, and Haibin Ling, "Clustered object detection in aerial images," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 8311–8320.
- [14] Changlin Li, Taojiannan Tao, Chen Gong, Xiaojun Chang, Liang-Chieh Chen, and Zizhang Wu, "Density map guided object detection in aerial images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 190–199.
- [15] Jinwang Wang, Chang Xu, Wen Yang, and Lei Yu, "A normalized gaussian Wasserstein distance for tiny object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2021.
- [16] Chang Xu, Jinwang Wang, Wen Yang, Huai Yu, Lei Yu, and Gui-Song Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *European Conference on Computer Vision*, 2022, pp. 526–543.
- [17] Chen Guo, Yang Li, Jiaman Ma, Zhenyu Fang, Zuoang Niu, and Hongbo Xu, "RESBIDET: Efficient dual-branch small object detection for UAVs under resource-constrained conditions," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 3156–3160.
- [18] Yixuan Li et al., "Adaptive sparse feature location activation strategy for sparse detectors on drone images," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [19] Shigeng Wang et al., "Foreground-enhanced coarse-to-fine detection for UAV small objects," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 1–5.
- [20] Kai Wang et al., "High-frequency detail compensation and multi-scale feature fusion net for UAV remote sensing object detection," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 1–5.
- [21] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [22] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2017, pp. 2980–2988.